USN | | | | | | | | | | [23CSBSBE11]

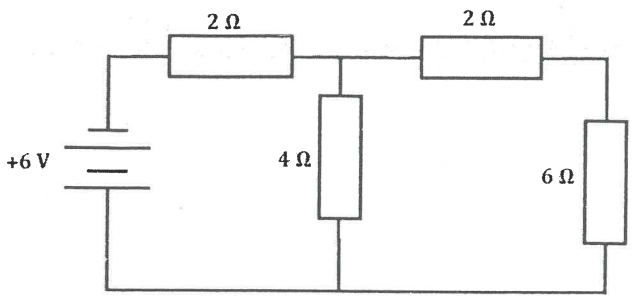
FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2025

PRINCIPLES OF ELECTRICAL ENGINEERING

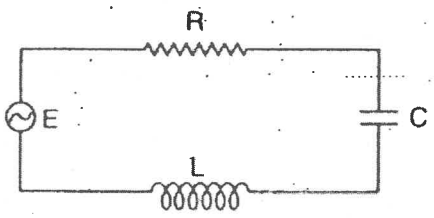
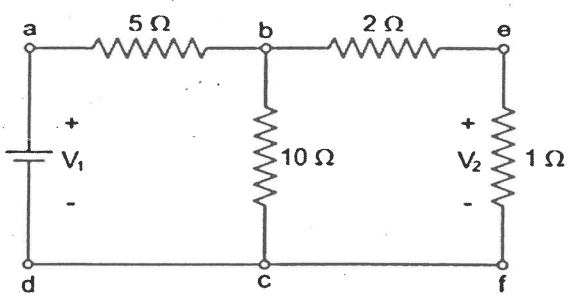
Time: 3 Hours **Maximum Marks: 100**

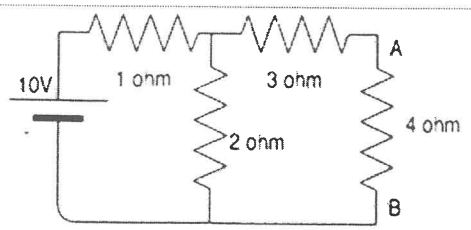
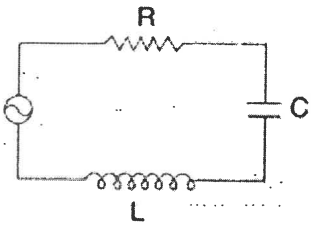
Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART - A	**5x2=10**		
1.	**a.**	What are active and passive elements?	L1	CO1	**02**
	b.	State Norton's theorem. What do you mean by Norton's equivalent circuit?	L1	CO2	**02**
	c.	An alternating voltage given by V = 140 sin 314 t is connected across a pure resistor of 50 Ω. Find the frequency of the source.	L3	CO3	**02**
	d.	A metallic rod of length l is rotated at a constant angular speed to, normal to a uniform magnetic field B. Derive an expression for the current induced in the rod, if the resistance of the rod is R.	L2	CO4	**02**
	e.	What is the Necessity of earthing?	L2	CO6	**02**
		PART - B	**6x5=30**		
2.	**a.**	Derive an expression for the impedance of an a.c. circuit consisting of an inductor and a resistor.	L2	CO2	**05**
		OR			
	b.	A voltage $V = V_0 \sin wt$ is applied to a series LCR circuit. Derive the expression for the average power dissipated over a cycle.	L2	CO2	**05**
3.	**a.**	Derive the expression for the self-inductance of a long solenoid of cross-sectional area A and length l, having n turns per unit length.	L2	CO4	**05**
		OR			
	b.	Define electric dipole moment. Write its S.I. unit. Write the expression for the work done on an electric dipole of dipole moment p in turning it from its position of stable equilibrium to a position of unstable equilibrium in a uniform electric field E.	L2	CO4	**05**
4.	**a.**	Calculate the current through the resistor of resistance 6 Ω by using Thevenin's theorem. 	L3	CO2	**05**

Deemed-to-be University Learning Resource Centre Jain Global Campus Ramanagara-562

		OR			
	b.	Can Thevenin's Theorem be applied to AC circuits? Give a Comparative discussion between Thevenin's Theorem with Norton's Theorem.	L2	CO2	**05**
5.	**a.**	Explain the underlying principle of a transformer. How is the large-scale transmission of electric energy over long distances done with the use of transformers?	L2	CO5	**05**
		OR			
	b.	Derive the expression of capacitance in parallel plate capacitors.	L4	CO4	**05**
6.	**a.**	Briefly explain the types of Earthing.	L2	CO6	**05**
		OR			
	b.	The figure shows a series LCR circuit connected to a variable frequency 200 V, 50 Hz source with L =100 mH, C = 100 μF and R = 100 Ω. Determine **i.** Current drawn **ii.** Power and Power Factor.  	L3	CO2	**05**
7.	**a.**	What is the importance of Ohm's Law? Determine the current in a 75 W light bulb that is connected to 120V.	L3	CO1	**05**
		OR			
	b.	Discuss about the independent and dependent sources. Also mention differences between ideal and practical sources.	L2	CO1	**05**
		PART - C	**6x10=60**		
8.	**a.**	What is Kirchhoff's voltage law? In the circuit shown below, determine (V2 / V1) using mesh method.  	L3	CO1	**10**
		OR			
	b.	State Kirchhoff's current law. Also mention the three significant limitations and applications of Kirchhoff's law.	L2	CO1	**10**
9.	**a.**	Find out the Thevenin resistance across the terminal AB for the circuit given below. Also, find out the Vth and current in the circuit.	L3	CO2	**10**

					
		OR			
	b.	The figure shows a series LCR circuit with L = 5.0 H, C = 80 μF, R = 40 Ω connected to a 240 V, source. Calculate **i.** The Power and Power Factor. **ii.** The current. **iii.** The rms potential drop across the capacitor, resistor and inductor. 	L3	CO2	**10**
10.	**a.**	Discuss the simplification of networks using Star/Delta transformation with an example. Mention differences between Star Circuit and Delta Circuit.	L2	CO3	**10**
		OR			
	b.	Draw and explain with the phasor diagram for a given AC Voltage applied to, **i.** Pure Resistor. **ii.** Pure Inductor. **iii.** Pure Capacitor.	L2	CO3	**10**
11.	**a.**	Mention various energy losses in a transformer and explain how they vary with load?	L2	CO5	**10**
		OR			
	b.	Discuss Faraday's law and Ampere's law.	L1	CO5	**10**
12.	**a.**	Briefly explain the important characteristics of Transducer. Also explain different sensors.	L2	CO6	**10**
		OR			
	b.	Briefly explain types of Wiring System &Wiring Accessories.	L2	CO6	**10**
13.	**a.**	Prove that an ideal capacitor in an a.c. circuit does not dissipate power.	L2	CO4	**10**
		OR			
	b.	Two charged conductors having capacitance C1 and C2 are of potentials V1 and V2 respectively are connected by a conducting wire. Calculate the common potential and loss of energy in connection.	L3	CO4	**10**